Dhruv Goyal

ep23btech11008@iith.ac.in | goyaldhruv942@gmail.com | +91 8800468029 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.57
J.K.G. International School	Class XII (CBSE)	2023	97.40%
J.K.G. International School	Class X (CBSE)	2021	99.40%

TECHNICAL ACHIEVEMENTS

Selected as a **Semifinalist in the NPCI Hackathon** for innovative fintech solutions

Participated in the Adobe Hackathon focused on real-world tech challenges

Specialist @Codeforces - Username: bigger_jigger

WORK EXPERIENCE

GroupBrain AI

AI Intern - Knowledge Graphs

- Developed a PDF-to-GraphRAG ingestion pipeline using Docling and canonical JSON, with rebuildable Neo4j graphs and Qdrant indexes preserving document, block, and chunk provenance.
- Integrated GLiNER/spaCy entity extraction with FastCoref and deterministic entity resolution to consolidate names, acronyms, and aliases while rejecting ambiguous matches within each document.
- Implemented GLiREL relation extraction with bounded candidate generation, resolved entity endpoints, and exact source-span validation to retain traceable relations and record rejection reasons.
- Built BGE-M3 dense/sparse retrieval with graph expansion and reranking, supported by index reconciliation, resumable ingestion, and scoped database operations for recoverable pipeline runs.

SKILLS

Programming languages: C/C++, Python, SQL, HTML/CSS/JS

Tools and Frameworks: Git, Github, Anaconda, Jupyter, MySQL, SOCKET.io, LaTeX, Solid Edge, LTspice

Technical Skills: Web Development, Object Oriented Programming, Basic Cryptography, DSA, Problem Solving

RELEVANT COURSES

Computer Science: Introduction to Programming, Introduction to AI, Data Structure, Computational Physics

Mathematics: Elementary Linear Equations, Differential Equations, Introduction to Probability, Introduction to Statistics, Maths for Physics, Advanced Maths for Physics, Number Theory, Non-Linear Dynamics

Electrical Engineering: Basic Electrical Circuits, Basic Electrical Engineering, Digital Circuits, Analog Electronics, Electronic Device Physics, Electronics Lab, Electrical Lab

Physics: Thermodynamics, Classical Mechanics, Modern Optics, Special Relativity, Electricity & Magnetism, Fluid Dynamics

PROJECTS

RewardTrade - Real-Time Loyalty Point Exchange

- Built a loyalty-point trading platform with merchant liquidity pools and peer-to-peer order books.
- Developed routing to split exchanges between fixed-rate pools and market-driven order books.
- Used Flask, Celery, Redis, and SQLAlchemy for backend services and asynchronous transaction processing.
- Designed APIs and merchant tools for account linking, point valuation, and transaction handling.

Strategic Election Manipulation in Multi-Agent Systems

- Modelled voter beliefs, knowledge, and decisions using epistemic logic under incomplete information.
- Developed game-theoretic strategies for adversarial agents to influence voter choices.
- Simulated multi-agent elections to examine effects on outcomes, fairness, and system robustness

Agentic RAG System with Memory and Observability

- Built a LangChain and LangGraph workflow to orchestrate retrieval and context-aware LLM responses.
- Integrated Qdrant semantic search to retrieve relevant passages as evidence for answer generation.
- Implemented Mem0 memory to retain user preferences and conversational context across sessions.
- Added Langfuse tracing to inspect agent execution, monitor LLM calls, and debug retrieval workflows.

Fraud Analytics

- Built AnoGAN and BiGAN anomaly detectors in PyTorch, trained exclusively on normal credit-card transactions.
- Scaled features with normal-only data and computed reconstruction and feature-matching anomaly scores.
- Implemented iterative latent optimization for AnoGAN and encoder-based reconstruction for BiGAN.
- Explored F1-based thresholds and compared accuracy, precision, recall, and ROC-AUC on labeled transactions.

GNN-QGST

- Represented quantum gate-set tomography circuits as graphs with gate, qubit, and measurement nodes linked by temporal and spatial edges.
- Developed a relational GCN with edge-specific message passing and probability-conditioned adaptive layer normalization to predict gate-noise parameters.
- Integrated a differentiable Pauli-transfer-matrix simulator to reconstruct circuit outcome probabilities from predicted noise maps.
- Implemented curriculum training and held-out evaluation against a Transformer baseline using MSE, KL divergence, total variation distance, and chi-square metrics.

NNQS-AIMD

- Built an H2 variational Monte Carlo workflow using JAX, DeepQMC, FermiNet, and Metropolis sampling.
- Implemented response-corrected force estimators with finite-difference checks and covariance analysis.
- Developed a JIT-compatible BAOAB/Langevin integrator for harmonic experiments with noisy forces.
- Prototyped differentiable electron space warping and documented force-estimator limitations and diagnostics.

EasyMoney

- Built a Python/C++20 paper-trading system to ingest Kite market depth, normalize ticks, and record L2 data.
- Integrated a C++ engine via pybind11 for order books, routing, order management, and risk checks.
- Combined mean-reversion and order-flow signals with exposure limits and transaction-cost filters.
- Added backtesting, walk-forward analysis, simulated fills, and SQLite logs of execution outcomes and latency.

Real-Time Chaotic Encryption via Henon Map

- Built a Python client-server application for real-time messaging over local networks using sockets.
- Implemented Henon-map chaotic encryption with per-character key updates to vary the encoding sequence.
- Integrated message encoding and decoding into the socket-based client-server messaging workflow.

Marine Robotics

- Modeled a nonlinear 4-DOF AUV with added mass, hydrodynamic damping, and body-to-earth kinematics.
- Implemented adaptive RBF neural-network sliding-mode controllers, including a double-loop PID design for trajectory tracking.
- Tested helix and sinusoidal paths under 15% model uncertainty, sensor noise, ocean currents, and thruster saturation.
- Compared three controllers using trajectory plots, per-axis tracking errors, and root mean squared error.

2D Ising Model

- Simulated a 2D spin lattice with Metropolis sampling and periodic boundary conditions.

- Computed magnetization, energy, and heat capacity across temperatures to examine phase transitions.
- Visualized energy distributions and spin configurations near the critical temperature.

Cancer Cell Detection - Fractal Morphology Analysis

- Built a Python image-analysis pipeline using OpenCV, SciPy, and scikit-image to study cell morphology.
- Implemented connected-component and marker-based watershed segmentation to isolate individual cell regions.
- Computed boundary and filled-cell fractal dimensions using Canny edges and vectorized box counting.
- Visualized segmentation masks and fractal-dimension distributions with per-image summary statistics.

POSITIONS OF RESPONSIBILITY

Competitions Head at Elan & nVision - The annual techno-cultural fest at IIT Hyderabad for 2025-26 tenure

- Spearheading the planning and execution of 25+ technical and non-technical nationwide competitions
- Collaborating with industry partners and internal departments to ensure smooth execution and maximum outreach

Internship Coordinator at OCS - the Office of Career Services, for the tenure 2025-26

- Facilitated internship opportunities by coordinating with companies across diverse sectors
- Acted as a liaison between students and recruiters, ensuring smooth execution of the internship drive

UG Mentor Sunshine - The Counselling Cell at IIT Hyderabad for the tenure 2024-25

Irshaad Domain Core of RangDeManch - The Dramatics club, for tenure 2024-25

Overall Coordinator at Elan & nVision - The annual techno-cultural fest at IIT Hyderabad for 2026-27 tenure:

- Directed the planning and execution of Elan & nVision, IIT Hyderabad's flagship three-day techno-cultural festival.
- Led cross-functional teams to deliver large-scale competitions, workshops, cultural performances and concerts.
- Managed festival finances, including budgeting, expense tracking, vendor payments and resource allocation.
- Coordinated with artists, sponsors, vendors and institute faculties to ensure smooth logistics and a great participant experience

Informals Coordinator at Elan & nVision - The annual techno-cultural fest at IIT Hyderabad for 2024-25 tenure

UG Buddy Sunshine - The Counselling Cell at IIT Hyderabad for the tenure 2023-24

EXTRACURRICULAR

Selected and currently part of **NSO (National Sports Organization)**

FIDE Chess Rating of 1534 in standard time format

Represented IIT Hyderabad at the **Inter IIT Cultural Meet 7.0** as a part of the dramatics team.

Winner of E-Sports Competition(Bullet Echo) organized at Elan & nVision 2025